Task 6A Rafael William 30/1/26

1 3 Algorithms for ~~cla~~ cache memory replacement.

→ Least Recently Used (LRU)

→ Replacing the cache block that hasn't been accessed for the longest period of time.

→ It uses the principle of <u>temporal locality</u>.

→ It work by making a system that keeps track of "use" timestamp and/or maintaining a stack of usage.

→ When a replacement is needed, the "oldest" entry is stashed.

→ First-in, First-out (FIFO)

→ Replacing the block that has been in cache the longest, no matter how often it used.

→ The system works like a queue.

→ The first block would eventually goes to the last slot in the cache.

→ When there's new entry, the first-in ~~item~~ will leave first.

→ Least Frequently Used (LFU)

→ This algorithm replace a block with the least amount of "clicks" since it was in the cache.

→ Each block will get a counter, when the cache is full, the least counter is evicted

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~~value is evicted~~.

2. Which algorithm is the most efficient.

→ Least ~~Recently~~ frequently Used (~~LRU~~) (LFU)

why ?

→ Higher cache hit percentage.

↳ in real-life use, the ~~block~~ LRU will experienced less "cachemiss"

↳ The more a data was used, the more possibility that it will be used again in short-time.

→ Exploit Temporal Usability

↳ The most used data has the possibility that it was used in a loop, or local variable.

↳ This "hot" data would be kept "warm" by ~~LRU~~ LFU algorithm.

→ Balance of performance

↳ While the FIFO is more simple and the LRU is more perfopmative

↳ The LFU algorithm has more balance in performance and material

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